3. Minimal surface

As an example of the framework presented here, the stiffness matrix, dynamic relaxation with viscous and kinetic damping [9] and force density methods are compared by form finding a hyperbolic paraboloid cable-net (Figure 1) with fixed boundaries. Three mesh sizes are computed, with an increasing number of degrees of freedom (3 per node): n=3x3, 27 dof's (Figure 1), n=7x7, 147 dof's and n=15x15, 675 dof's. The form finding starts from an initially flat net (Table 2).

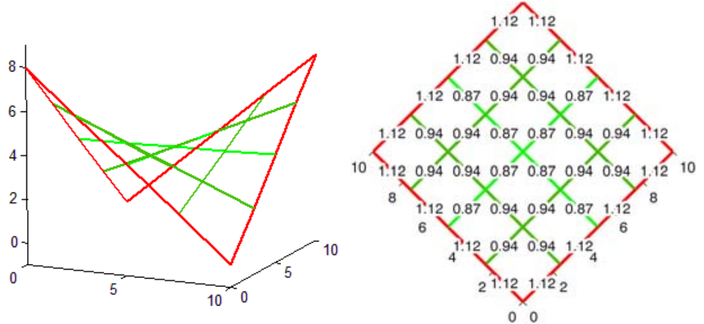

Fig. 1: Normalized forces $\mathbf{f}_{norm}$ *for* $Q=1$*, or* $EA=F_o=1$

As a first trial, trivial input parameters are used: Q=1, EA=F_0=1 and all unstressed lengths L_0 are equal. The solution is considered to have converged when the squared sum of all residual forces changes by less than 1% between iterations, or when the residual forces are 0. The forces in the results are normalized to their average for comparison ($\mathbf{f}_{\text{norm}}=\mathbf{f}/F_{\text{avg}}$).